

LIU HENGZHI

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EDUCATION

National University of Singapore <i>Bachelor of Computing in Computer Science (Tuition Grant)</i>	2026 – June 2029
Temasek Polytechnic, Singapore <i>Diploma in Biomedical Engineering with Merit</i> Awarded Polytechnic Scholarship, Course Bronze Medalist, and Director's List honors.	2023 – 2026

SKILLS

Programming Languages: C++, C, Python, SQL

Embedded & Hardware Engineering: Embedded Systems, Real-time Systems (FreeRTOS), ESP32, Firmware Development, PCB Design, Hardware-Software Co-Design

Tools & Utilities: CAD (Fusion 360, PTC Creo), Git/GitHub, Linux, FastAPI, MySQL, RESTful APIs, Adobe Suite

TECHNICAL PROJECTS

Smart Mahjong Play-Board <i>Independent Systems Developer</i> <i>C++, ESP32, PCB Design, Embedded Firmware, Serial Protocols</i> - Developed ESP32-based smart play-board hardware featuring custom PCB circuit integration. - Established low-latency hardware-software communication by designing customized serial protocols in embedded firmware.	2026
DTx Cognitive SaaS (ADHD Therapeutics) <i>Full-Stack Developer</i> <i>React, FastAPI, AWS Aurora, LLM Agents</i> - Built a clinical ADHD SaaS platform featuring adaptive test pacing calibrated to severity. - Reduced data latency by implementing local telemetry batching with AWS token authentication.	2026

WORK EXPERIENCE

National University of Singapore (iHealthtech) <i>Research Assistant Intern</i> - Generated high-quality experimental datasets for biomedical research by setting up microscale testing workflows. - Ensured data integrity across all testing pipelines by performing comprehensive validation checks on hardware instruments. - Improved data traceability by establishing structured digital documentation standards for all experimental records.	Mar 2025 – Jul 2025 <i>Singapore</i>
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CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing) Won 1st Place (TP Selection Round), demonstrating advanced CAD modeling and precision design.	2024
Electronic Design & Robotics Competitions Developed embedded logic systems and real-time control algorithms for robotics challenges.	2024